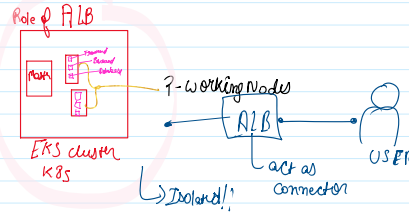
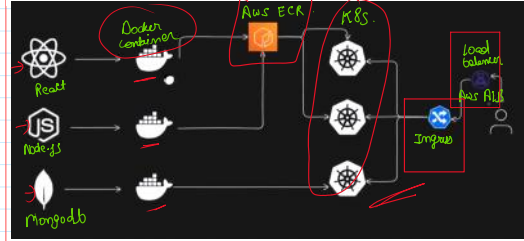


Architecture Overview



To encrypt/decrypt password -

```
ubuntu@ip-172-31-42-35:~/TWSThreeTierAppChallenge$ echo 'shubham' | base64
c2h1YmhhbQo=
ubuntu@ip-172-31-42-35:~/TWSThreeTierAppChallenge$ echo 'c2h1YmhhbQo=' | base64 --decode
shubham
```

What is headless service : where type is not given.

• Kubernetes ["Do installation at home directory"]

→ Creating cluster via EKS

• Kubectl install..

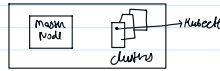
```
curl -o kubectl https://amazon-eks.s3.us-west-2.amazonaws.com/1.19.6/2021-01-05/linux/amd64/kubectl
chmod +x kubectl (to make it executable)
sudo mv kubectl /usr/local/bin (why we mv it in BIN → to get kubectl in environment path)
kubectl version --short --client
```

• EKSctl install.. → background silently download ✓

```
curl -s -o kubectl https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_$(uname -s)_amd64.tar.gz | tar xz -C /tmp
sudo mv /tmp/eksctl /usr/local/bin
eksctl version
```

Role of Kubelet "Controlling the cluster"

K8s "forming the cluster"



• Setting up EKS cluster

```
eksctl create cluster --name three-tier-cluster --region ap-south-1 --node-type t2.medium --nodes-min 2 --nodes-max 2
```

→ 2 clusters will be formed of t2.medium in ap-south-1 region

• deploying stack is forming on 'cloudFormation' in AWS

≈ takes 20 min

- forms control plane
- gateway
- deploying EKS Nodebook
- Roles form

• Linking Kubelet to EKS cluster

```
aws eks update-kubeconfig --region us-west-2 --name three-tier-cluster
```

→ Kubectl get nodes ✓ will run

⇒ 3 files { deploy.yaml ⇒ will make container/pod
• secrets.yaml ⇒ user name, password accessible.
• service.yaml ⇒ Makes deployment accessible. }

```
ubuntu@ip-172-31-42-35:~/TWSThreeTierAppChallenge/k8s_manifests/mongo$ kubectl create namespace workshop
namespace/workshop created
ubuntu@ip-172-31-42-35:~/TWSThreeTierAppChallenge/k8s_manifests/mongo$ kubectl apply -f deploy.yaml
```

```
ubuntu@ip-172-31-42-35:~/TWSThreeTierAppChallenge/k8s_manifests/mongo$ kubectl apply -f secrets.yaml
secret/mongo-sec created
ubuntu@ip-172-31-42-35:~/TWSThreeTierAppChallenge/k8s_manifests/mongo$ kubectl get pod
NAME READY STATUS RESTARTS AGE
mongodb-797968fbc5-m2hfw 1/1 Running 0 4213s
ubuntu@ip-172-31-42-35:~/TWSThreeTierAppChallenge/k8s_manifests/mongo$
```

```
ubuntu@ip-172-31-13-232:~/TWSThreeTierAppChallenge/kubernetes-manifests-file/backend$ kubectl get pods -n three-tier
NAME READY STATUS RESTARTS AGE
api-56967b5796-gk2p 1/1 Running 0 4213h
mongodb-154dc18b-rtfm 1/1 Running 0 4213h
ubuntu@ip-172-31-13-232:~/TWSThreeTierAppChallenge/kubernetes-manifests-file/backend$ kubectl logs api-56967b5796-gk2p -n three-tier
Listening on port 3500...
Connected to database
ubuntu@ip-172-31-13-232:~/TWSThreeTierAppChallenge/kubernetes-manifests-file/backend$
```

↳ Same can also be done by a Docker but why Kubernetes is used
↳ for auto scaling
Rebuilding if instance hampered.